Bridge Material Type and Structural Condition: A Statewide Analysis of Florida's Bridge Inventory

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Introduction

Choosing the correct bridge material is an essential engineering decision made at the foundations of the construction process, normally during design time. Nevertheless, long-term effects on structures due to specific regional environments are often generalized rather than statistically examined and quantified. In Florida, the humid and coastal climate tends to accelerate deterioration processes, such as corrosion. However, there has not been enough state-wide inspection data comparing the relative resilience of common bridge materials.

The research objective is to determine the relationship between bridge material type and structural condition rating for bridges in the state of Florida.

Methods

For this research, the 2025 Federal Highway Administration National Bridge Inventory (NBI) dataset was used to obtain information on structural condition ratings as well as material classifications for a total of 11,680 bridges. Originally, the dataset included 13,070 bridges, but 1,390 were excluded due to insufficient data or structural design categories not comparable to the four primary material categories: concrete, prestressed concrete, steel, and steel continuous. Bridges were grouped by material type and their overall structural condition, rated as “Good”, “Fair”, or “Poor” per NBI methodology, was compared across groups. To determine whether the material type is associated with the structural condition rating, a chi-square test of independence was used.

Results

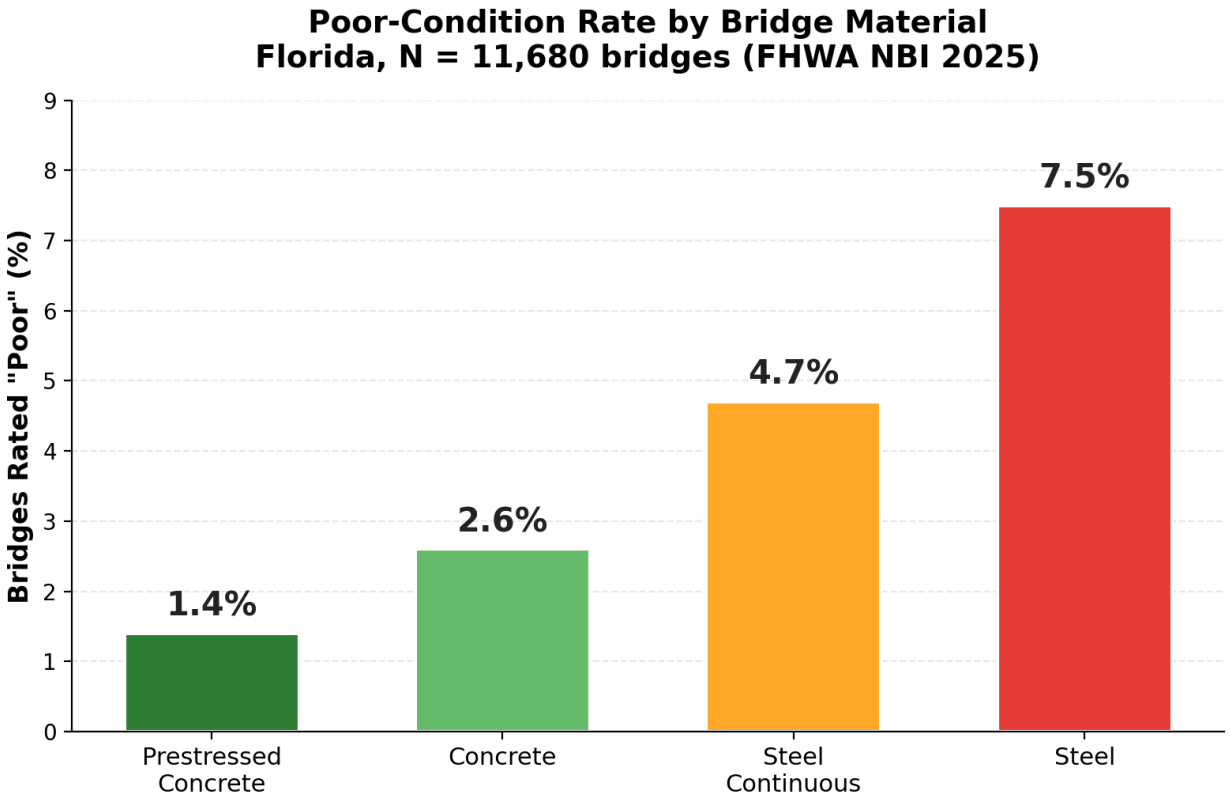
For each group category, the number of bridges analyzed (n) was as follows: 6,454 Prestressed Concrete, 3,779 Concrete, 845 Steel, and 602 Steel Continuous. For each group, the structural condition distribution was: for Concrete, 1,821 were Fair, 1,859 were Good, and 99 were Poor; for Prestressed Concrete, 2,029 were Fair, 4,336 were Good, and 89 were Poor; for Steel, 394 were Fair, 388 were Good, and 63 were Poor; for Steel Continuous, 204 were Fair, 370 were Good, and 28 were Poor. These values indicate that Steel bridges contain the highest rate of Poor condition (7.5%), followed by Steel Continuous (4.7%), Concrete (2.6%), and Prestressed Concrete with the lowest Poor-condition rate (1.4%). Furthermore, Prestressed Concrete bridges also have the highest rate of Good condition (67.2%), while for Steel bridges it was only 45.9%.

After completing a chi-square test of independence, the “distance” score obtained was $\chi^2 = 485.26$, with degrees of freedom (df) of 6, and a probability value (p) of less than 0.001. These values indicate that the differences between materials are highly unlikely to be coincidental.

Discussion

Analyzing the results, it is possible to conclude that bridge material type is indeed significantly related to structural conditions across the bridges in Florida. As seen above, steel bridges are much more likely to be found in poor conditions than concrete or prestressed concrete bridges. This result makes sense when taking into consideration the fact that Florida contains a chloride-rich, humid coastal environment. Prior materials engineering research has established that chloride exposure accelerates steel corrosion, but it was rarely quantified at a statewide statistical scale. This research uses statistics based on a comprehensive federal inspection data to contribute to a statewide confirmation of the relationship between the environment and materials, rather than isolated and generalized case studies.

It is important to note that this study did not consider other variables, by way of illustration, bridge age, average daily traffic, or maintenance history, which may independently affect the condition ratings. Future work could potentially use multivariate regression to examine these factors and isolate their effects on material type with higher precision. Nonetheless, the findings in this research still offer engineers in humid climate regions data that can be helpful when selecting materials for their construction. Studying materials is an essential step that needs to be taken so final products can be more resilient in similar environments.



Chi-square test of independence: $\chi^2(6, N=11{,}680) = 485.26$, $p < 0.001$